

François-Pierre Paty

Applied mathematics · Machine learning · Optimal transport · Time series · Operations research

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Applied mathematician (PhD, École Polytechnique engineer) working at the intersection of machine learning, optimization and operations research. I built the entire R&D of an AI startup (Ida, acquired by RELEX Solutions) from scratch, published at top machine-learning venues (ICML, AISTATS) on algorithmic optimal transport, and run a Chinese-learning platform and newsletter that reach more than 25,000 people worldwide.

EXPERIENCE

Sept 2025

– present

Visiting Professor of Mathematics

École Centrale de Pékin (Beihang University) · Beijing, China

- Teaching mathematics (analysis, probability, differential calculus and equations) in French to Chinese students: lectures, tutorials and oral examinations (khôlles).
- Distilling and presenting technical material to varied audiences, and collaborating in a multicultural, Chinese-speaking environment.

Oct 2023

– Sept 2025

Research Scientist & R&D Lead (first employee)

Ida (acquired by RELEX Solutions in 2025, now RELEX Fresh) · Paris, France

- Startup optimizing fresh-food orders for grocery retailers. Built the entire R&D pipeline from scratch: research, implementation, production deployment and fast iteration in close contact with field data and data engineers.
- Impact: 30–50% reduction in food waste, 3 margin points gained, deployment across hundreds of stores in Europe.
- Modeling: time-series forecasting (ML), stock reliability (HMM), order optimization (MILP, MDP, dynamic programming).
- Production: deployment and monitoring in Python, Git and CI/CD.
- Mentored MSc-level data-science interns; led technical presentations to clients and in tenders.

May 2022

– present

Founder & creator

HSK Level & Chinese Grammar Gems · Worldwide

- Chinese-proficiency testing platform (120,000+ tests, 5M answers) and a Substack Bestseller newsletter with 25,000+ subscribers worldwide.
- Full-stack development, algorithmic modeling and AI-assisted app building.
- Growing recurring revenue (+50% in 2025); building a new app integrating generative AI.

Sept 2021

– April 2022

Postdoctoral Researcher

CREST, CNRS · Palaiseau, France

- Developed novel machine-learning algorithms with applications in economics; optimal transport applied to labor-market modeling, with numerical simulations in Python (JAX).

Sept 2018

– Aug 2021

PhD Researcher & Teaching Assistant

ENSAE Paris, Institut Polytechnique de Paris · Palaiseau, France

- Algorithmic optimal transport under the supervision of Marco Cuturi (CREST funding).
- 3 publications (ICML ×2, AISTATS) including a Notable Paper Award; 13 talks at international conferences and seminars.
- Teaching assistant in mathematics, optimization, machine learning and deep learning (BSc and MSc levels).

April 2017
– Aug 2017

Research Intern, Applied Mathematics

EDF R&D & Finance for Energy Market Research Centre · Palaiseau, France

- Adapted sparse-deconvolution techniques to missing-data imputation for financial time series. Received the congratulations of the Applied Mathematics department of École polytechnique; a patent was filed by EDF (FR3075969B1).

EDUCATION

2018
– 2021

PhD in Applied Mathematics and Machine Learning

Institut Polytechnique de Paris (ENSAE) · Palaiseau, France

Thesis: “Optimal transport in high dimension: obtaining regularity and robustness using convexity and projections”, advised by Marco Cuturi.

2017
– 2018

MSc in Statistics and Machine Learning (Mathématiques de l'Aléatoire, M2)

Université Paris-Sud (Paris-Saclay) · Orsay, France

Advanced courses in theoretical statistics and machine learning.

2017
– 2018

Engineering degree (Data Science track)

ENSAE Paris · Palaiseau, France

Specialized in statistics and data science.

2014
– 2018

Ingénieur polytechnicien (X2014)

École Polytechnique · Palaiseau, France

Applied mathematics, with a focus on statistics, probability and data analysis.

2012
– 2014

Classe préparatoire (MP)

Lycée Louis-le-Grand · Paris, France

Intensive mathematics and physics, preparing for the competitive entrance examinations to the Grandes Écoles.

PUBLICATIONS

5. **Algorithms for Weak Optimal Transport with an Application to Economics**
F-P. Paty, P. Choné, F. Kramarz · *Preprint 2022*
4. **Optimal transport in high dimension: obtaining regularity and robustness using convexity and projections**
F-P. Paty · *PhD thesis, Institut Polytechnique de Paris 2021*
3. **Regularized Optimal Transport is Ground Cost Adversarial**
F-P. Paty, M. Cuturi · *International Conference on Machine Learning (ICML) 2020*
2. **Regularity as Regularization: Smooth and Strongly Convex Brenier Potentials in Optimal Transport**
F-P. Paty, A. d'Aspremont, M. Cuturi · *International Conference on Artificial Intelligence and Statistics (AISTATS) 2020* · **Notable Paper Award (top 3 of 423 accepted papers)**
1. **Subspace Robust Wasserstein Distances**
F-P. Paty, M. Cuturi · *International Conference on Machine Learning (ICML) 2019* · **Oral presentation (top 20% of accepted papers)**

AWARDS & DISTINCTIONS

- **2020** · Notable Paper Award (top 3 of 423 accepted papers), International Conference on Artificial Intelligence and Statistics (AISTATS).
- **2019** · Oral presentation (top 20% of accepted papers), International Conference on Machine Learning (ICML).
- **2017** · Congratulations of the Applied Mathematics department, École polytechnique.

TALKS, TUTORIALS & CONFERENCES

<i>Conferences</i>	• AISTATS 2020 — contributed talk (online), Aug 2020. • ICML 2020 — contributed talk (online), Jul 2020. • ICML 2019 — 20-minute oral presentation, Long Beach, Jun 2019.
<i>Seminars</i>	• New Monge Problems seminar, Université Gustave Eiffel, Jun 2022. • Mokaplan team seminar, INRIA Paris, May 2022. • Image, Optimization and Probability seminar, Institut de Mathématiques de Bordeaux, Mar 2021. • EDMH PhD students seminar, Université Paris-Saclay, Mar 2021. • Learning meets Astrophysics seminar day, CEA Saclay, Jan 2020. • Stat-Eco-ML seminar, ENSAE Paris, Nov 2019. • Le Séminaire Palaisien, INRIA Saclay, Nov 2019.
<i>Summer schools</i>	• Computational optimal transport tutorial, Machine Learning Summer School (MLSS), Moscow, Aug 2019. • Saint-Flour Probability Summer School, Jul 2019.
<i>Workshops</i> (poster / participant)	• Paris workshop on optimal transport with applications to economics and statistics, Sciences Po Paris, Oct 2021. • Poster, NeurIPS Optimal Transport and Machine Learning Workshop, Vancouver, Dec 2019. • People in Optimal Transportation and Applications, Cortona, Jun 2019. • Poster, Optimization and Statistical Learning workshop, Les Houches, Mar 2019.

TEACHING

2025 – 2026	École Centrale de Pékin (Beihang University) • Lectures (cours magistraux, in French) in advanced analysis and probability. • Tutorials in analysis, differential calculus and differential equations. • Oral examinations (khôlles).
Dec 2024	École Nationale des Ponts et Chaussées • Invited course on operations research.
2018 – 2021	ENSAE Paris — Teaching Assistant • Mathematics (in French): Topology and Functional Analysis (BSc); Differentiable Optimization (BSc); Mathematical Statistics (MSc). • Computer science (in English): Geometric Methods in Machine Learning (MSc); Stochastic Optimization and Automatic Differentiation for ML (MSc); Optimal Transport (MSc); Deep Learning: Models and Optimization (MSc).

SERVICE TO THE COMMUNITY

- Conference reviewer: AISTATS 2020, ICML 2020, NeurIPS 2020.
- Co-organizer of the Statistics, Econometrics, Machine Learning (Stat-Eco-ML) seminar, ENSAE Paris, 2019–2021.

SKILLS & LANGUAGES

Machine learning	Time series and forecasting, optimal transport, probabilistic models (HMM), deep learning, LLMs and generative AI.
Optimization & OR	MILP, dynamic programming, MDPs, operations research.
Programming Languages	Python (JAX, scikit-learn, PyTorch, cvxpy), SQL, Git, CI/CD, React, Claude Code.
	French (Native) · English (Fluent) · Italian (Conversational) · Chinese (HSK5)